

Tuples --> Immutable (17-08-26)

```
In [1]: t=()  
t
```

```
Out[1]: ()
```

```
In [2]: type(t)
```

```
Out[2]: tuple
```

```
In [3]: t1=(10,True,2.3,1+2j,"welcome")  
t1
```

```
Out[3]: (10, True, 2.3, (1+2j), 'welcome')
```

```
In [5]: t1.append(50) #tuple has no inbuilt function append  
t1
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[5], line 1  
----> 1 t1.append(50) #tuple has no inbuilt function append  
      2 t1  
  
AttributeError: 'tuple' object has no attribute 'append'
```

```
In [6]: t1
```

```
Out[6]: (10, True, 2.3, (1+2j), 'welcome')
```

```
In [8]: t1[0]
```

```
Out[8]: 10
```

```
In [9]: t1[0:4]
```

```
Out[9]: (10, True, 2.3, (1+2j))
```

```
In [10]: t1[0]=100  
t1
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[10], line 1  
----> 1 t1[0]=100  
      2 t1  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [13]: t1
```

```
Out[13]: (10, True, 2.3, (1+2j), 'welcome')
```

```
In [14]: t1.count(10)
```

Out[14]: 1

```
In [15]: t1.index(1+2j)
```

Out[15]: 3

```
In [16]: t=(10,20,30)
t
```

Out[16]: (10, 20, 30)

```
In [17]: t*3
```

Out[17]: (10, 20, 30, 10, 20, 30, 10, 20, 30)

Tuple practice

Tuple creation

```
In [2]: tup1=()
tup1
```

Out[2]: ()

```
In [3]: tup2=(10,20,30)
tup2
```

Out[3]: (10, 20, 30)

```
In [4]: tup3=(10.77,30.66,60.89)
tup3
```

Out[4]: (10.77, 30.66, 60.89)

```
In [5]: tup4=("one","two","three")
tup4
```

Out[5]: ('one', 'two', 'three')

```
In [6]: tup5=("Mousumi",25,(50,10),(150,90)) #Nested tuple
tup5
```

Out[6]: ('Mousumi', 25, (50, 10), (150, 90))

```
In [7]: tup6=(100,"Mousumi",17.89)
tup6
```

Out[7]: (100, 'Mousumi', 17.89)

```
In [8]: tup7=("Mousumi",25,(50,10),(150,90) , {"Raj","Shyam"},(99,22,33))
tup7
```

Out[8]: ('Mousumi', 25, (50, 10), (150, 90), {'Raj', 'Shyam'}, (99, 22, 33))

```
In [9]: len(tup7)
```

```
Out[9]: 6
```

Tuple Indexing

```
In [10]: tup2[0]
```

```
Out[10]: 10
```

```
In [11]: tup2[-1]
```

```
Out[11]: 30
```

```
In [15]: tup7[3][0]
```

```
Out[15]: 150
```

```
In [16]: tup4[0][0]
```

```
Out[16]: 'o'
```

```
In [17]: tup5[-1]
```

```
Out[17]: (150, 90)
```

Tuple slicing

```
In [19]: tup5
```

```
Out[19]: ('Mousumi', 25, (50, 10), (150, 90))
```

```
In [18]: tup5[::-1]
```

```
Out[18]: ((150, 90), (50, 10), 25, 'Mousumi')
```

```
In [41]: mytup=('one','two','three','four','five','six','seven','eight')  
mytup
```

```
Out[41]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [42]: mytup[0:3]
```

```
Out[42]: ('one', 'two', 'three')
```

```
In [43]: mytup[2:5]
```

```
Out[43]: ('three', 'four', 'five')
```

```
In [44]: mytup[:3]
```

```
Out[44]: ('one', 'two', 'three')
```

```
In [45]: mytup[:2]
```

```
Out[45]: ('one', 'two')
```

```
In [46]: mytup[-3:]
```

```
Out[46]: ('six', 'seven', 'eight')
```

```
In [47]: mytup[:-2]
```

```
Out[47]: ('one', 'two', 'three', 'four', 'five', 'six')
```

```
In [48]: mytup[-2:]
```

```
Out[48]: ('seven', 'eight')
```

```
In [49]: mytup[-1]
```

```
Out[49]: 'eight'
```

```
In [50]: mytup[:]
```

```
Out[50]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [51]: mytup[::-1]
```

```
Out[51]: ('eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one')
```

```
In [52]: mytup[::-2]
```

```
Out[52]: ('eight', 'six', 'four', 'two')
```

```
In [53]: mytup[::2]
```

```
Out[53]: ('one', 'three', 'five', 'seven')
```

```
In [54]: mytup[::5]
```

```
Out[54]: ('one', 'six')
```

Remove and change items

```
In [55]: mytup
```

```
Out[55]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [58]: del mytup[0] #Tuple are IMMUTABLE which means we can't DELETE tuple items
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[58], line 1
----> 1 del mytup[0] #Tupke are IMMUTABLE  which means we can't DELETE tuple item
s

TypeError: 'tuple' object doesn't support item deletion

```

In [59]: `mytup[0]=1 #Tuple are IMMUTABLE which means we can't CHANGE tuple items`

```

-----
TypeError                                Traceback (most recent call last)
Cell In[59], line 1
----> 1 mytup[0]=1 #Tuple are IMMUTABLE  which means we can't CHANGE tuple items

TypeError: 'tuple' object does not support item assignment

```

In [62]: `del mytup #Deleting emtire tuple object is possible`

```

-----
NameError                                Traceback (most recent call last)
Cell In[62], line 1
----> 1 del mytup #Deleting emtire tuple object is possible

NameError: name 'mytup' is not defined

```

In [63]: `mytup`

```

-----
NameError                                Traceback (most recent call last)
Cell In[63], line 1
----> 1 mytup

NameError: name 'mytup' is not defined

```

Index position

In [64]: `mytup=('one','two','three','four','five','six','seven','eight')`
`mytup`

Out[64]: `('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')`

In [65]: `mytup.index('one')`

Out[65]: `0`

In [66]: `mytup.index('five')`

Out[66]: `4`

Sorting

In [68]: `mytup1=(43,67,99,12,6,90,67)`
`mytup1`

Out[68]: `(43, 67, 99, 12, 6, 90, 67)`

```
In [70]: sorted(mytup1) #Returns new sorted list and doesn't change original tuple
```

```
Out[70]: [6, 12, 43, 67, 67, 90, 99]
```

```
In [71]: sorted(mytup1 , reverse=True) #Sort in desc order
```

```
Out[71]: [99, 90, 67, 67, 43, 12, 6]
```

Tuple Membership

```
In [72]: mytup
```

```
Out[72]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [73]: 'one' in mytup
```

```
Out[73]: True
```

```
In [74]: 'nine' in mytup
```

```
Out[74]: False
```

Loop through Tuple

```
In [75]: mytup
```

```
Out[75]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [76]: for i in mytup:  
         print(i)
```

```
one  
two  
three  
four  
five  
six  
seven  
eight
```

```
In [77]: mytup1
```

```
Out[77]: (43, 67, 99, 12, 6, 90, 67)
```

```
In [78]: for i in mytup1:  
         print(i)
```

```
43  
67  
99  
12  
6  
90  
67
```

In []: